Gramática del analizador sintáctico de Lua 5.5

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Notación

Los no terminales se escriben así: ⟨NON_TERMINAL⟩ y los terminales en minúsculas y sin llaves triangulares. El símbolo inicial de la gramática es es ⟨CHUNK⟩. ε es la cadena vacía, $\to$ introduce una regla y | separa sus alternativas. La gramática es LL(1).

1. Estructura del programa

Símbolo inicial, bloques y la sentencia `return`.

⟨CHUNK⟩ → ⟨BLOCK⟩

⟨BLOCK⟩ → ⟨STATLIST⟩ ⟨RETSTAT⟩

⟨STATLIST⟩ → ⟨STAT⟩ ⟨STATLIST⟩ | ε

⟨RETSTAT⟩ → `return` ⟨MAYBE_EXPLIST⟩ ⟨OPT_SEMILOCON⟩ | ε

⟨OPT_SEMILOCON⟩ → `tkn_semicolon` | ε

2. Sentencias

Las asignaciones y las llamadas se factorizan después del primer nombre.

⟨STAT⟩ → `tkn_semicolon`
| ⟨NAME⟩ ⟨VAR_STAT⟩
| ⟨LABEL⟩
| `break`
| `goto` ⟨NAME⟩
| `while` ⟨EXP⟩ ⟨DO_BLOCK_END⟩
| `repeat` ⟨BLOCK⟩ `until` ⟨EXP⟩
| ⟨IFSTAT⟩
| ⟨FORSTAT⟩
| `function` ⟨FUNCNAME⟩ ⟨FUNCBODY⟩
| ⟨LOCAL_DEC⟩
| ⟨GLOBAL_DEC⟩

⟨VAR_STAT⟩ → `tkn_opening_bra` ⟨EXP⟩ `tkn_closing_bra` ⟨VAR_STAT⟩
| `tkn_period` ⟨NAME⟩ ⟨VAR_STAT⟩
| ⟨FUNCTIONCALL⟩ ⟨FUNCALL_STAT⟩
| ⟨MORE_VAR⟩ `tkn_assign` ⟨EXPLIST⟩

⟨FUNCALL_STAT⟩ → `tkn_opening_bra` ⟨EXP⟩ `tkn_closing_bra` ⟨VAR_STAT⟩

$$
\begin{aligned}
&\quad | \texttt{tkn_period}\ \langle \text{NAME} \rangle\ \langle \text{VAR_STAT} \rangle \\
&\quad | \langle \text{FUNCTIONCALL} \rangle\ \langle \text{FUNCALL_STAT} \rangle \ \mid\ \varepsilon \\
\langle \text{LABEL} \rangle &\rightarrow \texttt{tkn_goto}\ \langle \text{NAME} \rangle\ \texttt{tkn_goto} \\
\langle \text{DO_BLOCK_END} \rangle &\rightarrow \texttt{do}\ \langle \text{BLOCK} \rangle\ \texttt{end} \\
\langle \text{IFSTAT} \rangle &\rightarrow \texttt{if}\ \langle \text{EXP} \rangle\ \texttt{then}\ \langle \text{BLOCK} \rangle\ \langle \text{ELSEIFSTAT} \rangle\ \langle \text{ELSESTAT} \rangle\ \texttt{end} \\
\langle \text{ELSEIFSTAT} \rangle &\rightarrow \texttt{elseif}\ \langle \text{EXP} \rangle\ \texttt{then}\ \langle \text{BLOCK} \rangle\ \langle \text{ELSEIFSTAT} \rangle \ \mid\ \varepsilon \\
\langle \text{ELSESTAT} \rangle &\rightarrow \texttt{else}\ \langle \text{BLOCK} \rangle \ \mid\ \varepsilon \\
\langle \text{FORSTAT} \rangle &\rightarrow \texttt{for}\ \langle \text{NAME} \rangle\ \langle \text{FORTYPE} \rangle\ \langle \text{DO_BLOCK_END} \rangle \\
\langle \text{FORTYPE} \rangle &\rightarrow \langle \text{NUMERICAL_FOR} \rangle \mid \langle \text{GENERIC_FOR} \rangle \\
\langle \text{NUMERICAL_FOR} \rangle &\rightarrow \texttt{tkn_assign}\ \langle \text{EXP} \rangle\ \texttt{tkn_comma}\ \langle \text{EXP} \rangle\ \langle \text{OPT_EXP} \rangle \\
\langle \text{OPT_EXP} \rangle &\rightarrow \texttt{tkn_comma}\ \langle \text{EXP} \rangle \ \mid\ \varepsilon \\
\langle \text{GENERIC_FOR} \rangle &\rightarrow \langle \text{MORE_NAMES} \rangle\ \texttt{in}\ \langle \text{EXPLIST} \rangle
\end{aligned}
$$

3. Declaraciones

`local` y `global`.

$$
\begin{aligned}
\langle \text{LOCAL_DEC} \rangle &\rightarrow \texttt{local}\ \langle \text{LOCAL_OPTS} \rangle \\
\langle \text{LOCAL_OPTS} \rangle &\rightarrow \langle \text{FUNC_DEC} \rangle \mid \langle \text{ATTNAME_BRANCH} \rangle \\
\langle \text{ATTNAME_BRANCH} \rangle &\rightarrow \langle \text{ATTNAMELIST} \rangle\ \langle \text{OPT_EXPLIST} \rangle \\
\langle \text{GLOBAL_DEC} \rangle &\rightarrow \texttt{global}\ \langle \text{GLOBAL_OPTS} \rangle \\
\langle \text{GLOBAL_OPTS} \rangle &\rightarrow \langle \text{FUNC_DEC} \rangle \mid \langle \text{ATTNAMELIST} \rangle\ \langle \text{OPT_EXPLIST} \rangle \\
\langle \text{FUNC_DEC} \rangle &\rightarrow \texttt{function}\ \langle \text{NAME} \rangle\ \langle \text{FUNCBODY} \rangle \\
\langle \text{OPT_EXPLIST} \rangle &\rightarrow \texttt{tkn_assign}\ \langle \text{EXPLIST} \rangle \ \mid\ \varepsilon \\
\langle \text{ATTNAMELIST} \rangle &\rightarrow \langle \text{NAME} \rangle\ \langle \text{ATTRIB} \rangle\ \langle \text{NAME_ATTRIB_LIST} \rangle \\
\langle \text{NAME_ATTRIB_LIST} \rangle &\rightarrow \texttt{tkn_comma}\ \langle \text{NAME} \rangle\ \langle \text{ATTRIB} \rangle\ \langle \text{NAME_ATTRIB_LIST} \rangle \ \mid\ \varepsilon \\
\langle \text{ATTRIB} \rangle &\rightarrow \texttt{tkn_less}\ \langle \text{NAME} \rangle\ \texttt{tkn_greater} \ \mid\ \varepsilon
\end{aligned}
$$

4. Funciones

Nombres de función, cuerpos y listas de parámetros.

$$
\begin{aligned}
\langle \text{FUNCNAME} \rangle &\rightarrow \langle \text{NAME} \rangle\ \langle \text{DOTTED_FUNCNAME} \rangle\ \langle \text{COLON_FUNCNAME} \rangle \\
\langle \text{DOTTED_FUNCNAME} \rangle &\rightarrow \texttt{tkn_period}\ \langle \text{NAME} \rangle\ \langle \text{DOTTED_FUNCNAME} \rangle \ \mid\ \varepsilon
\end{aligned}
$$

⟨COLON_FUNCNAME⟩ → `tkn_colon` ⟨NAME⟩ | ε

⟨FUNCTIONDEF⟩ → `function` ⟨FUNCBODY⟩

⟨FUNCBODY⟩ → `tkn_opening_par` ⟨OPT_PARLIST⟩ `tkn_closing_par` ⟨BLOCK⟩ `end`

⟨OPT_PARLIST⟩ → ⟨PARLIST⟩ | ε

⟨PARLIST⟩ → ⟨NAME⟩ ⟨PARLIST_TAIL⟩ | ⟨VARPARAGRAM⟩

⟨PARLIST_TAIL⟩ → `tkn_comma` ⟨PARLIST_AFTER_COMMA⟩ | ε

⟨PARLIST_AFTER_COMMA⟩ → ⟨NAME⟩ ⟨PARLIST_TAIL⟩ | ⟨VARPARAGRAM⟩

⟨VARPARAGRAM⟩ → `tkn_varargs` ⟨OPT_NAME⟩

⟨OPT_NAME⟩ → ⟨NAME⟩ | ε

5. Listas

Listas de variables, de nombres y de expresiones.

⟨MORE_NAMES⟩ → `tkn_comma` ⟨NAME⟩ ⟨MORE_NAMES⟩ | ε

⟨MORE_VAR⟩ → `tkn_comma` ⟨PREFIXEXP⟩ ⟨MORE_VAR⟩ | ε

⟨MAYBE_EXPLIST⟩ → ⟨EXPLIST⟩ | ε

⟨EXPLIST⟩ → ⟨EXPLIST_ITEM⟩ ⟨MORE_EXP⟩

⟨MORE_EXP⟩ → `tkn_comma` ⟨EXPLIST_ITEM⟩ ⟨MORE_EXP⟩ | ε

⟨EXPLIST_ITEM⟩ → ⟨EXP⟩ | ⟨FUNCTIONDEF⟩

6. Expresiones

Operandos, expresiones prefijo y llamadas.

⟨EXP⟩ → ⟨OPERAND⟩ ⟨EXP_PRIME⟩

⟨EXP_PRIME⟩ → ⟨ARITH_TAIL⟩ ⟨CMP_TAIL⟩ ⟨LOGIC_TAIL⟩

⟨OPERAND⟩ → ⟨NAME⟩ ⟨PREFIXEXP_PRIME⟩ | ⟨OPERAND_NOT_ID⟩

⟨OPERAND_NOT_ID⟩ → `nil` | `false` | `true`
| `tkn_num` | `tkn_str` | ⟨TABLECONSTRUCTOR⟩
| ⟨UNOP⟩ ⟨OPERAND⟩ | `tkn_opening_par` ⟨EXP⟩ `tkn_closing_par`

⟨PREFIXEXP⟩ → ⟨NAME⟩ ⟨PREFIXEXP_PRIME⟩

⟨PREFIXEXP_PRIME⟩ → `tkn_opening_bra` ⟨EXP⟩ `tkn_closing_bra` ⟨PREFIXEXP_PRIME⟩
| `tkn_period` ⟨NAME⟩ ⟨PREFIXEXP_PRIME⟩
| ⟨FUNCTIONCALL⟩ ⟨PREFIXEXP_PRIME⟩ | ε

⟨FUNCTIONCALL⟩ → ⟨ARGS⟩ | `tkn_colon` ⟨NAME⟩ ⟨ARGS⟩

⟨ARGS⟩ → `tkn_opening_par` ⟨MAYBE_EXPLIST⟩ `tkn_closing_par`

7. Tablas

Constructores de tablas.

⟨TABLECONSTRUCTOR⟩ → `tkn_opening_key` ⟨FIELDLIST⟩ `tkn_closing_key`

⟨FIELDLIST⟩ → ⟨FIELD⟩ ⟨FIELD_TAIL⟩ | ε

⟨FIELD_TAIL⟩ → ⟨FIELDSEP⟩ ⟨FIELDLIST⟩ | ε

⟨FIELD⟩ → `tkn_opening_bra` ⟨EXP⟩ `tkn_closing_bra` `tkn_assign` ⟨FIELD_VALUE⟩
| ⟨NAME⟩ ⟨FIELD_ID⟩
| ⟨OPERAND_NOT_ID⟩ ⟨EXP_PRIME⟩
| ⟨FUNCTIONDEF⟩

⟨FIELD_ID⟩ → `tkn_assign` ⟨FIELD_VALUE⟩
| ⟨PREFIXEXP_PRIME⟩ ⟨EXP_PRIME⟩

⟨FIELD_VALUE⟩ → ⟨EXP⟩ | ⟨FUNCTIONDEF⟩

⟨FIELDSEP⟩ → `tkn_comma` | `tkn_semicolon`

8. Operadores

Operadores binarios por nivel, operadores unarios y nombres.

⟨LOGIC_TAIL⟩ → ⟨LOGICAL_OP⟩ ⟨CMP_EXP⟩ ⟨LOGIC_TAIL⟩ | ε

⟨CMP_EXP⟩ → ⟨ARTH_EXP⟩ ⟨CMP_TAIL⟩

⟨CMP_TAIL⟩ → ⟨RELATIONAL_OP⟩ ⟨ARTH_EXP⟩ | ε

⟨ARTH_EXP⟩ → ⟨OPERAND⟩ ⟨ARITH_TAIL⟩

⟨ARITH_TAIL⟩ → ⟨FLAT_OP⟩ ⟨OPERAND⟩ ⟨ARITH_TAIL⟩ | ε

⟨FLAT_OP⟩ → ⟨ARITH_OP⟩ | ⟨BITWISE_OP⟩ | ⟨OTHER_OP⟩

⟨ARITH_OP⟩ → `tkn_plus` | `tkn_minus` | `tkn_times` | `tkn_div`
| `tkn_floor_div` | `tkn_power` | `tkn_mod`

⟨RELATIONAL_OP⟩ → `tkn_less` | `tkn_leq` | `tkn_greater`
| `tkn_geq` | `tkn_equal` | `tkn_neq`

⟨LOGICAL_OP⟩ → `and` | `or`

⟨BITWISE_OP⟩ → `tkn_bit_and` | `tkn_bitex_or` | `tkn_bit_or`
| `tkn_right_shift` | `tkn_left_shift`

⟨OTHER_OP⟩ → `tkn_concat`

⟨UNOP⟩ → `tkn_minus` | `not` | `tkn_length`

⟨NAME⟩ → `id` | `print` | `pcall`
| `error` | `warn` | `xpcall`